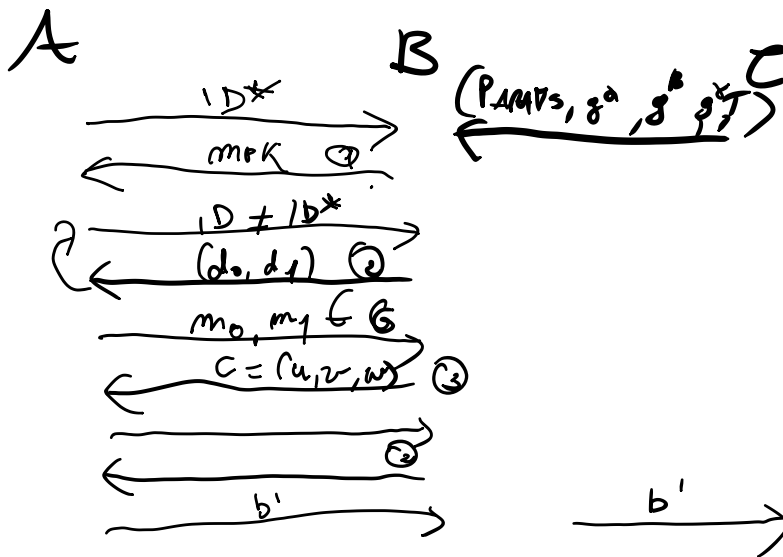


Cryptography - 16_12

Proof

Let $\mathcal{A}$ be a PPT adversary breaking selective IND-ID-CPA of Π . We construct PPT $\mathcal{B}$ breaking Decisional Bilinear Diffie Hellman (DBDH) assumption.



Note: $g_1 = g^\alpha, g_2 = g^\beta, g_3 = g^\gamma$.

Let's start with (1).

The $mpk = (params, g_1, g_2, h)$. Where $g_1 = g^\alpha$ and $g_2 = g^\beta$.

We'll have that:

$$h = g_1^{-ID^*} \cdot g^a \in \mathbb{G}; a \leftarrow \mathbb{Z}_q$$

Usually h is uniform.

(Note that $msk = g_2^\alpha = g^{\alpha\beta}$ is unknown to $\mathcal{B}$).

Also, recall:

$$F(ID) = g_1^{ID} \cdot h$$

Let's take a look at (2). Extraction queries, with $ID \neq ID^*$.

We'll reply with:

$$d_{ID} = (d_0, d_1)$$

where $d_0 = g_2^{\frac{-a}{ID-ID^*}} \cdot F(ID)^r$, with a being the value picked before, and

$$d_1 = g^{\tilde{r}} = g^{r - \frac{\beta}{ID - ID^*}} = \frac{g^r}{g_2^{\frac{1}{ID - ID^*}}}$$

(

The original extraction was:

- $d_1 = g^r; d_0 = g_2^\alpha \cdot F(ID)^r$
-)

Here's step (3). We have the Challenge $ctx: m_0, m_1 \in \mathbb{G}$.

$$c = (u, v, w)$$

(

Recall the original:

- $u = \hat{e}(g_1, g_2)^\gamma$
- $v = g^\gamma$
- $w = F(ID^*)^\gamma$
-)

$$v = y^\gamma; u = T \cdot m_b; w = g_3^a$$

Analysis:

1. The mpk is perfectly simulated. Note $h = g_1^{-ID^*} \cdot g^a$ is uniform.

3)The challenge ctx

Missing part

$$w = g_3^a$$

Note that $F(ID^*)^\gamma = (g_1^{ID^*} \cdot h)^\gamma = (g_1^{ID^*} \cdot g_1^{-ID^*} \cdot g^a)^\gamma = g_3^a$

2. Extraction queries $ID \neq ID^*$.

$$d_1 = g^{\tilde{r}} \quad \text{where } \tilde{r} = r - \frac{\beta}{ID - ID^*}$$

This is good as $\tilde{r}$ is still uniform.

We must prove that

$$d_0 = g_2^\alpha \cdot F(ID)^{\tilde{r}}$$

Well, recall:

$$d_0 = g_2^{\frac{-\alpha}{ID - ID^*}} \cdot F(ID)^r$$

$$\begin{aligned}
&= g_2^{\frac{-a}{ID-ID^*}} \cdot (g_1^{ID} \cdot \underbrace{(g^{-ID^*} \cdot g^a)}_h)^r = \\
&= g_2^{\frac{-a}{ID-ID^*}} \cdot (g_1^{ID-ID^*} \cdot g^a)^r = \\
&= (g_1^{ID-ID^*} \cdot g^a)^{\frac{\beta}{ID-ID^*}} \cdot \frac{(g_1^{ID-ID^*} \cdot g^a)^r \cdot g_2^{r - \frac{a}{ID-ID^*}}}{(g_1^{ID-ID^*} \cdot g^a)^{\frac{\beta}{ID-ID^*}}} = \\
&= g_1^\beta \cdot g^{a \cdot \frac{\beta}{ID-ID^*}} \cdot g_2^{\frac{-a}{ID-ID^*}} \cdot (g_1^{ID-ID^*} \cdot g^a)^{r - \frac{\beta}{ID-ID^*}} = \\
&= \underbrace{g_2^\alpha}_{msk} \cdot \underbrace{(g_1^{ID-ID^*} \cdot g^a)^{\tilde{r}}}_{F(ID)}
\end{aligned}$$

■

Now, let's see some applications, especially two of them.

1. CCA-secure PKE
2. UF-CMA signatures.

Let's start with the first. We use two ingredients:

1. $\Pi = (Setup, KGen, Enc, Dec)$ an IBE scheme.
2. $\Pi' = (KGen', Sign, Vrfy)$ a signature scheme.

Define $\Pi'' = (KGen'', Enc'', Dec'')$.

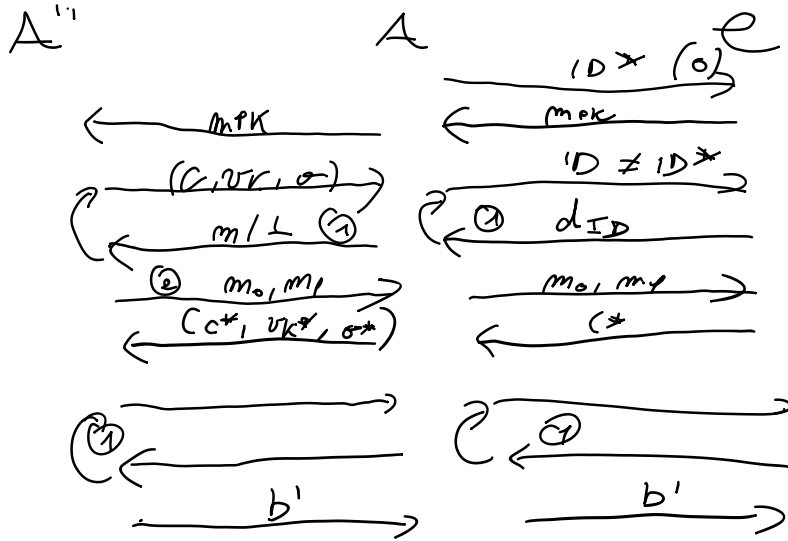
- $KGen''(1^\lambda)$: $(mpk, msk) \leftarrow Setup(1^\lambda)$ and output ek (encryption key)
 $ek = mpk$; $dk = msk$.
- $Enc''(ek, m)$: Get $(vk, sk) \leftarrow KGen'(1^\lambda)$ such that $|vk| = n(\lambda)$. Output $c'' = (c, vk, \sigma)$, where $c \leftarrow Enc(mpk, ID = vk, m)$, and $\sigma \leftarrow Sign(sk, c)$.
- $Dec''(dk, c'')$. Given $c'' = (c, vk, \sigma)$ if $Vrfy(vk, c, \sigma) = 0$ output $\perp$. Else, output $Dec(mpk, d_{vk}, c)$ where $d_{vk} \leftarrow KGen(msk, vk = ID)$.

✓ Theorem

Assuming Π is selective IND-ID-CPA and Π' is strongly UF-CMA, then Π'' is CCA-secure.

We will basically give a sketch.

Assume $\exists$ PPT $\mathcal{A}''$ breaking Π'' . Consider the following reduction to Π .



We have $vk^* = ID^*$.

Given a query (c, vk, σ) . First, if $Vrfy(vk, c, \sigma) = 0$, output $\perp$. Else:

- If $vk \neq vk^*$, then send $vk = ID$ to the challenger, receive d_{ID} and output $Dec(mpk, d_{id}, c)$.
- If $vk = vk^*$, we can't send $ID^* = vk^*$ to $\mathcal{C}$. We will reject this query. Notice that $(c, \sigma) \neq (c^*, \sigma^*)$, otherwise $(c, vk, \sigma) = (c^*, vk^*, \sigma^*)$. Then, we can just reject this query.

2. Upon $m_0, m_1 \in \mathbb{G}$, then send m_0, m_1 to $\mathcal{C}$, obtaining

$$c^* \leftarrow Enc(mpk, ID^* = vk^*, m_b). \text{ Output } (c^*, vk^*, \sigma^*) \leftarrow Sign(sk, c)$$

The actual proof:

1. Create a hybrid where decryption queries (c, vk, σ) with $vk = vk^*$ but $(c, \sigma) \neq (c^*, \sigma^*)$ are rejected. By UFCMA, this is indistinguishable.
2. Use the above reduction to IND-ID-CPA to show that no attacker can break CCA security in the hybrid.

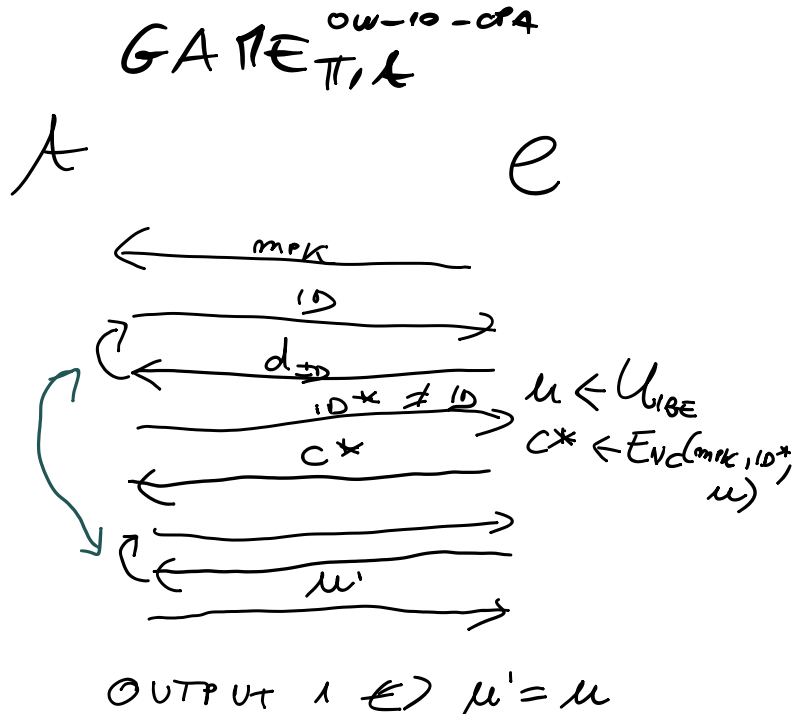
We finish up by showing that IBE also implies signatures.

Let $\Pi = (Setup, KGen, Enc, Dec)$ be an IBE. Consider $\Pi' = (KGen', Sign, Vrfy)$.

- $KGen(1^\lambda)'$: $(mpk, msk) \leftarrow Setup(1^\lambda)$, output $vk = mpk$ and $sk = msk$.
- $Sign(sk, m)$: Think of $m = ID$, output $\sigma = d_m \leftarrow KGen(msk, m)$.

- $\text{Vrfy}(vk, m, \sigma)$: Again think of $m = ID$ and $\sigma = d_{ID}$. Pick $\mu \leftarrow \mathcal{U}_{IBE}$ (plaintext for IBE), and let $c \leftarrow \text{Enc}(mpk, ID = m, \mu)$. Then accept $(m, \sigma) \iff \text{Dec}(d_{ID}, c) = \mu$.

We'll show UFCMA security follows from the following weak property of IBE: OW-ID-CPA.



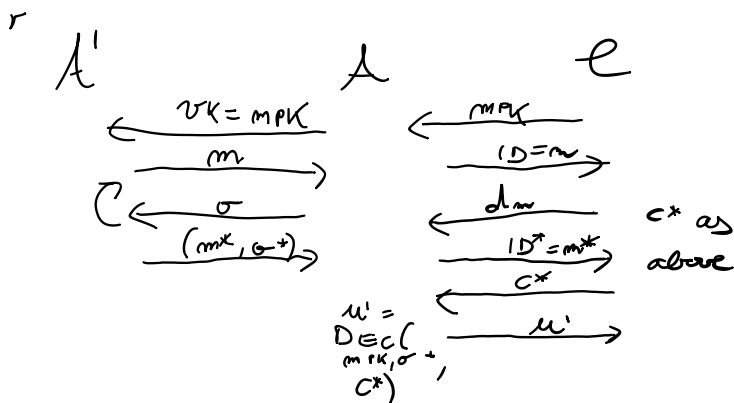
We are going to need OW-ID-CPA with negligible advantage: this is implied by IND-ID-CPA + $\frac{1}{|\mathcal{U}_{IBE}|} = \text{negl}(\lambda)$.

✓ Theorem

Assuming Π satisfies OW-ID-CPA, then Π' is UF-CMA.

📖 Proof

Simple reduction. Assume $\exists$ PPT $\mathcal{A}'$ against UFCMA, we build PPT $\mathcal{A}$ against the IBE.



$$Pr[\mathcal{A} \text{ wins}] \geq Pr[\mathcal{A}' \text{ wins}] \geq \frac{1}{poly(\lambda)}$$

■

NOTE: in the GAME shown above, the queries before the sending of μ' are not needed.

The reduction would in principle break in case the IBE is only selectively secure.

$\implies$ We need IBE to be IND-ID-CPA.

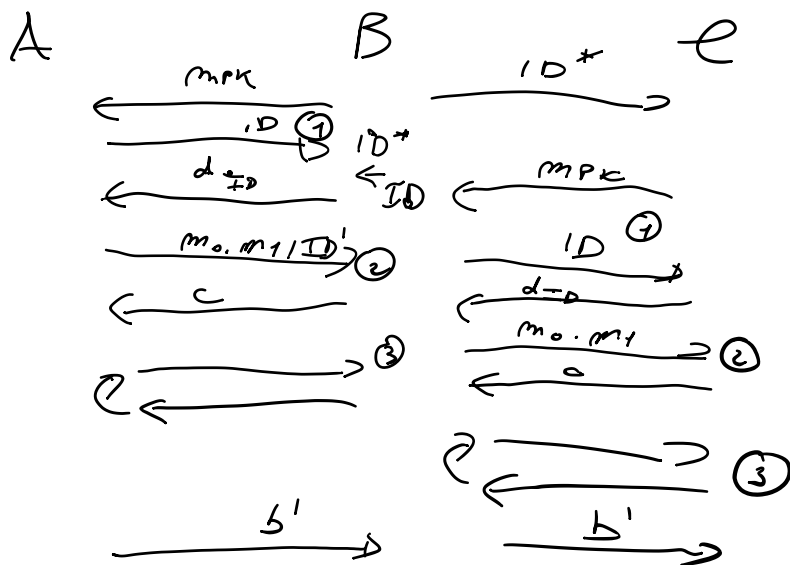
The good news: While there are direct constructions, it is also true that any selective IND-ID-CPA IBE is also IND-ID-CPA with a certain loss in the parameters:

✓ Theorem

Any IBE that is $(t, q, \mathcal{E})$ -selectively IND-ID-CPA is also $(t, q, N\mathcal{E})$ -IND-ID-CPA with $N = |\mathcal{ID}|$ (identity space).

🕒 Proof sketch

Given $\mathcal{A}$ breaking IND-ID-CPA build $\mathcal{B}$ breaking the selective version.



(1) If $ID = ID^*$, abort. Else, we are good.

(2) If $ID' = ID^*$ abort. Else, we are good.

(3) Assuming we did not abort earlier, $ID \neq ID^*$ and we are fine.

Call q_1 the number of queries of kind (1), and q is the total number of queries. t is the running time.

$$Pr[\neg ABORT] \geq \left(1 - \frac{q_1}{N}\right) \cdot \frac{1}{N - q_1} = \frac{1}{N}$$

$\implies$ The distinguisher's advantage is $\geq \frac{\epsilon}{N}$.